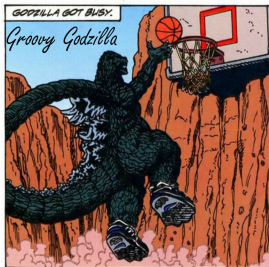


Predicting March Madness Basketball Games

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Introduction

- March Madness is the NCAA Division I Basketball Tournament, a single-elimination bracket of 68 teams held each March and April across both the Men's and Women's divisions to crown the National Champions.
- Teams earn a spot either by winning their conference tournament (an automatic bid) or by being selected by the NCAA Selection Committee (an at-large bid).
- The regular season determines the seeding. Teams are seeded 1 through 16 within each of four regions, where lower seeds are considered the strongest and are rewarded with matchups against the highest seeds.
- With 63 games decided by single elimination, upsets are frequent, which is why the tournament has become famous for its unpredictability, hence the name "March Madness."

Introduction

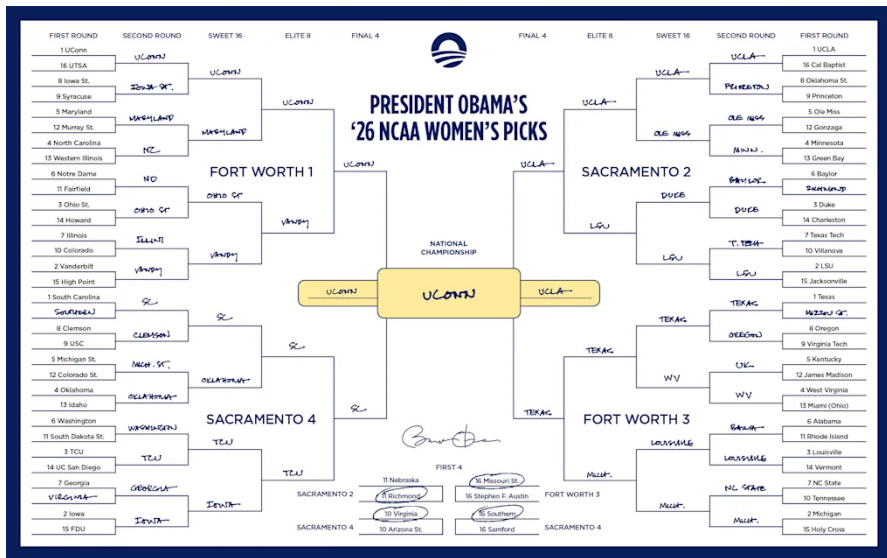


Figure 1: 2026 Bracket Example

- Kaggle is an online platform for data science and machine learning competitions, owned by Google.
- Companies, researchers, and organizations post prediction problems with datasets, evaluation metrics, and prize pools.
- Participants submit models that are scored on a hidden test set, with a public leaderboard ranking submissions in real time.
- Kaggle has become a benchmark in the ML community for evaluating predictive modeling approaches against a global field of competitors.

March Machine Learning Mania

- Annual Kaggle competition running since 2014, challenging participants to predict outcomes of every possible matchup in the NCAA tournament.
- Submissions are a probability $P \in [0, 1]$ for each of the $2 \times \binom{68}{2} = 4556$ possible matchups across Men's and Women's. Scored using the Brier

Score:

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2 \quad (1)$$

- Where f_t is our prediction and o_t is the outcome.
- o_t is 1 if the team wins, and 0 if they lose.

Prediction Difficulty

- Top predictive models plateau near 75% accuracy.
- Odds of a perfect bracket: 1 in 9.2 quintillion (random) to 1 in 128 billion (informed).
- No publicly verified bracket has survived past the Sweet Sixteen.
- \$3.1 billion wagered in 2025, sportsbook lines are the benchmark to beat.